



2026 年 01 月 CSCA 考试数学

January 2026 CSCA Exam Mathematics

题目(Questions)、答案(Answers)与解析(Explanations)

第 1 题

题目(Question): 设集合 $A = \{6, 7, 8, 9, 10\}$, Φ 表示空集, 则下列说法正确的是? (Let set $A = \{6, 7, 8, 9, 10\}$, and Φ denote the empty set. Then which of the following statements is correct?)

A $9 \in A$ (9 belongs to A)

B $\{6\} \in A$ ($\{6\}$ belongs to A)

C $6 \subseteq A$ (6 is a subset of A)

D $\Phi \in A$ (Φ belongs to A)

答案(Answer): A

解析(Explanation):

- 选项 A: 9 是集合 A 的元素, 元素与集合的“属于”关系(符号 $\in$) 成立, 故 A 正确。(For option A: 9 is an element of set A, and the “membership” relation (symbol $\in$) between elements and sets holds, so A is correct.)
- 选项 B: $\{6\}$ 是集合, 集合与集合间应是“包含”关系($\subseteq$), 而非“属于”($\in$), 故 B 错误。(For option B: $\{6\}$ is a set, and the relation between sets is “inclusion” (symbol $\subseteq$) rather than “membership” ($\in$), so B is incorrect.)
- 选项 C: 6 是集合 A 的元素, 元素与集合间不能用“包含”关系($\subseteq$), 应使用“属于”($\in$), 故 C 错误。(For option C: 6 is an element of set A, and the “inclusion” relation ($\subseteq$) cannot be used between elements and sets; instead, “membership” ($\in$) should be used, so C is incorrect.)

- 选项 D: 空集 Φ 不包含在集合 A 中, 故 $\Phi \notin A$, D 错误。(For option D: The empty set Φ is not contained in set A, so $\Phi \notin A$, and D is incorrect.)

第 2 题

题目(Question): 设集合 $A = \{x \mid -2 \leq x \leq 2\}$, $B = \{x \mid x > 0\}$, 则 $A \cup B = ?$ (Let sets $A = \{x \mid -2 \leq x \leq 2\}$ and $B = \{x \mid x > 0\}$. Then $A \cup B = ?$)

A $\{x \mid x \leq -2\}$

B $\{x \mid 0 < x \leq 2\}$

C $\{x \mid x \geq -2\}$

D $\{x \mid -2 \leq x < 0\}$

答案(Answer): C

解析(Explanation):

- 集合的并集 $A \cup B$ 表示所有属于 A 或属于 B 的元素组成的集合。(The union $A \cup B$ of sets denotes the set composed of all elements belonging to A or B.)
- 集合 A 覆盖 $-2 \leq x \leq 2$, 集合 B 覆盖 $x > 0$, 合并后覆盖范围为 $x \geq -2$, 故 $A \cup B = \{x \mid x \geq -2\}$, C 正确。(Set A covers $-2 \leq x \leq 2$, and set B covers $x > 0$. The combined range is $x \geq -2$, so $A \cup B = \{x \mid x \geq -2\}$, and C is correct.)

第 3 题

题目(Question): 不等式 $x^2 - x - 2 > 0$ 的解集是? (What is the solution set of the inequality $x^2 - x - 2 > 0$?)

A $\{x \mid x < -1 \text{ 或 } x > 2\}$

B $\{x \mid -1 < x < 2\}$

C $\{x \mid x < -2 \text{ 或 } x > -1\}$



$$D \{x \mid -2 < x < 1\}$$

答案(Answer):A

解析(Explanation):

- 对二次函数因式分解: $x^2 - x - 2 = (x - 2)(x + 1)$ 。(First, factorize the quadratic function: $x^2 - x - 2 = (x - 2)(x + 1)$.)
- 不等式 $(x - 2)(x + 1) > 0$ 表示两因式同号。(The inequality $(x - 2)(x + 1) > 0$ means the two factors have the same sign.)
 - 当 $x - 2 > 0$ 且 $x + 1 > 0$ 时, 解得 $x > 2$; (When $x - 2 > 0$ and $x + 1 > 0$, the solution is $x > 2$;)
 - 当 $x - 2 < 0$ 且 $x + 1 < 0$ 时, 解得 $x < -1$ 。(When $x - 2 < 0$ and $x + 1 < 0$, the solution is $x < -1$.)
- 综上, 解集为 $\{x \mid x < -1 \text{ 或 } x > 2\}$, A 正确。(In summary, the solution set is $\{x \mid x < -1 \text{ or } x > 2\}$, and A is correct.)

第 4 题

题目(Question): 设 $\{a_n\}$ 是等差数列(arithmetic sequence), 首项 $a_1 = 2$, 公差 $d = 3$, 则

$a_{100} = ?$ (Let $\{a_n\}$ be an arithmetic sequence with the first term $a_1 = 2$ and common difference $d = 3$. Then $a_{100} = ?$)

A 299

B 302

C 305

D 298

答案(Answer):A

解析(Explanation):

- 等差数列通项公式: $a_n = a_1 + (n-1)d$ 。 (The general term formula of an arithmetic sequence is: $a_n = a_1 + (n-1)d$.)
- 代入 $a_1 = 2$ 、 $d = 3$ 、 $n = 100$:(Substitute $a_1 = 2$, $d = 3$, $n = 100$:)
$$a_{100} = 2 + (100-1) \times 3 = 2 + 297 = 299$$
- 故 A 正确。 (Thus, A is correct.)

第 5 题

题目(Question): 函数 $f(x) = \frac{1}{x} + \sqrt{1-x}$ 的定义域是? (What is the domain of the function

$$f(x) = \frac{1}{x} + \sqrt{1-x} \text{ ?})$$

A $(-\infty, 1]$

B $(-\infty, 0) \cup (0, 1)$

C $(-\infty, 0) \cup (0, 1]$

D $[1, +\infty) \cup [1, +\infty)$

答案(Answer): C

解析(Explanation):

- 函数定义域需满足所有表达式有意义: (The domain of a function must satisfy that all expressions are meaningful.)
 1. 分式 $\frac{1}{x}$ 有意义的条件是 $x \neq 0$; (The fraction $\frac{1}{x}$ is meaningful if and only if $x \neq 0$;)
 2. 二次根式 $\sqrt{1-x}$ 有意义的条件是 $1-x \geq 0$, 即 $x \leq 1$ 。 (The square root $\sqrt{1-x}$ is meaningful if and only if $1-x \geq 0$, i.e., $x \leq 1$.)
- 综合得定义域为 $(-\infty, 0) \cup (0, 1]$, C 正确。 (Combining the two conditions, the domain is $(-\infty, 0) \cup (0, 1]$, and C is correct.)



第 6 题

题目 (Question): 在平面直角坐标系 (rectangular coordinate system) 中, 点 $P(1,2)$ 与点 Q 关于 x 轴对称(symmetrical about the x -axis), 则点 Q 的坐标是? (In the rectangular coordinate system, point $P(1,2)$ is symmetrical about the x -axis with point Q . What are the coordinates of point Q ?)

A: (2, 1)

B: (1, -2)

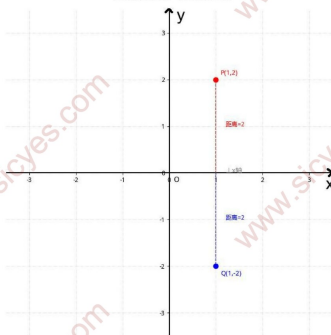
C: (-1, 2)

D: (-1, -2)

答案 (Answer): B**解析 (Explanation):**

- 关于 x 轴对称的点的性质: 横坐标(abscissa)不变, 纵坐标(ordinate)互为相反数。(Property of points symmetrical about the x -axis: the abscissa remains unchanged, and the ordinate is the opposite number.)
- 点 $P(1,2)$ 的横坐标为 1, 纵坐标为 2, 故对称点 Q 的坐标为 (1, -2), B 正确。(The abscissa of point $P(1,2)$ is 1 and the ordinate is 2, so the coordinates of the symmetrical point Q are (1, -2), and B is correct.)

平面直角坐标系 - 点的对称关系



第 7 题

题目(Question): 设角 $\alpha = 30^\circ$, 则下列说法正确的是? (Let angle $\alpha = 30^\circ$. Then which of the following statements is correct?)

A $\sin \alpha = \frac{\sqrt{2}}{2}$

B $\cos \alpha = \frac{\sqrt{3}}{2}$

C $\cos \alpha = \frac{1}{2}$

D $\tan \alpha = \sqrt{3}$

答案(Answer):B

解析(Explanation):

- 特殊角三角函数值:(Trigonometric function values of special angles)
 - $\sin 30^\circ = \frac{1}{2}$, 故 A 错误; ($\sin 30^\circ = \frac{1}{2}$, so A is incorrect;)
 - $\cos 30^\circ = \frac{\sqrt{3}}{2}$, 故 B 正确、C 错误; ($\cos 30^\circ = \frac{\sqrt{3}}{2}$, so B is correct and C is incorrect;)
 - $\tan 30^\circ = \frac{\sqrt{3}}{3}$, 故 D 错误。($\tan 30^\circ = \frac{\sqrt{3}}{3}$, so D is incorrect.)

第 8 题

题目(Question): 关于函数 $f(x) = x^4 + 3$, 下列说法正确的是? (About the function $f(x) = x^4 + 3$, which of the following statements is correct?)

A 该函数是奇函数(The function is odd)

B 以上都不正确(None of the above)

C 该函数既不是奇函数也不是偶函数(The function is neither odd nor even)

D 该函数是偶函数(The function is even)

答案(Answer):D

解析(Explanation):



- 奇偶函数定义:(Definition of odd and even functions:)
 - 偶函数:对定义域内任意 x , 满足 $f(-x) = f(x)$; (Even function: For any x in the domain, $f(-x) = f(x)$ holds.)
 - 奇函数:对定义域内任意 x , 满足 $f(-x) = -f(x)$ 。 (Odd function: For any x in the domain, $f(-x) = -f(x)$ holds.)
- 函数 $f(x) = x^4 + 3$ 的定义域为 $\mathbb{R}$, 且 $f(-x) = (-x)^4 + 3 = x^4 + 3 = f(x)$, 故为偶函数, D 正确。(The domain of $f(x) = x^4 + 3$ is $\mathbb{R}$, and $f(-x) = (-x)^4 + 3 = x^4 + 3 = f(x)$, so it is an even function, and D is correct.)

第 9 题

题目(Question):函数 $y = x^3 + 3$ ($x \in \mathbb{R}$) 的反函数(inverse function)是? (What is the inverse function of the function $y = x^3 + 3$ ($x \in \mathbb{R}$)?)

A $y = \sqrt[3]{x-3}$, $x \in \mathbb{R}$

B $y = \sqrt[3]{x+3}$, $x \geq -3$

C $y = \sqrt[3]{x+3}$, $x \in \mathbb{R}$

D $y = \sqrt[3]{x-3}$, $x \geq 3$

答案(Answer):A

解析(Explanation):

- 求反函数步骤:(Steps to find the inverse function:)
 1. 由原函数解出 x : $x^3 = y - 3$, 即 $x = \sqrt[3]{y-3}$; (Solve for x from the original function: $x^3 = y - 3$, i.e., $x = \sqrt[3]{y-3}$;)
 2. 交换 x 和 y , 得反函数: $y = \sqrt[3]{x-3}$; (Swap x and y to get the inverse function: $y = \sqrt[3]{x-3}$;)

3. 原函数定义域为 $\mathbb{R}$ 、值域为 $\mathbb{R}$ ，故反函数定义域为 $\mathbb{R}$ 。(The domain and range of the original function are both $\mathbb{R}$, so the domain of the inverse function is $\mathbb{R}$.)

- 综上，反函数为 $y = \sqrt[3]{x-3}$ ($x \in \mathbb{R}$)，A 正确。(In summary, the inverse function is $y = \sqrt[3]{x-3}$ ($x \in \mathbb{R}$), and A is correct.)

第 10 题

题目(Question): 下列哪个点在第三象限(third quadrant)? (Which of the following points is in the third quadrant?)

A (-1, 2)

B (-1, -2)

C (1, 2)

D (1, -2)

答案(Answer): B

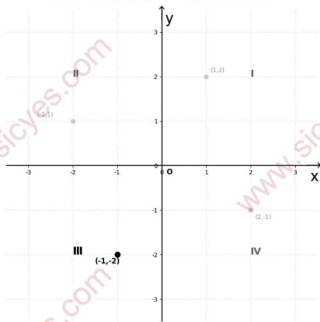
解析(Explanation):

- 平面直角坐标系象限定

义:(Definition of quadrants in the rectangular coordinate system:)

- 第一象限: $x > 0, y > 0$; (First quadrant: $x > 0, y > 0$;))
- 第二象限: $x < 0, y > 0$; (Second quadrant: $x < 0, y > 0$;))
- 第三象限: $x < 0, y < 0$; (Third quadrant: $x < 0, y < 0$;))

Cartesian Coordinate System - Features of Quadrant III





○ 第四象限: $x > 0, y < 0$ 。(Fourth quadrant: $x > 0, y < 0$.)

- 选项 B 中 $x = -1 < 0, y = -2 < 0$, 符合第三象限条件, 故 B 正确。(In option B, $x = -1 < 0$ and $y = -2 < 0$, which satisfies the condition of the third quadrant, so B is correct.)

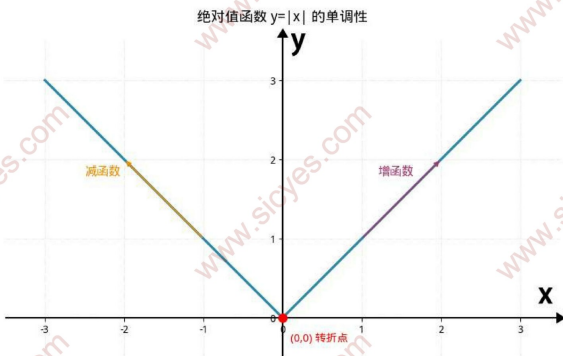
第 11 题

题目(Question): 已知 $y = |x|$, 下列结论正确的是? (Given $y = |x|$, which of the following conclusions is correct?)

- A 当 $x < 0$ 时, 函数是增函数(When $x < 0$, the function is an increasing function)
- B 当 $x > 0$ 时, 函数是增函数(When $x > 0$, the function is an increasing function)
- C 当 $x \in \mathbb{R}$ 时, 函数是减函数(When $x \in \mathbb{R}$, the function is a decreasing function)
- D 当 $x \in \mathbb{R}$ 时, 函数是增函数(When $x \in \mathbb{R}$, the function is an increasing function)

答案(Answer): B

解析(Explanation):



- 绝对值函数分段表达式: (Piecewise expression of the absolute value function $y = |x|$:)

$$y = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

- 单调性分析:(Analysis of function monotonicity;)
 - 当 $x > 0$ 时, $y = x$, 斜率为 $1 > 0$, 是增函数;(When $x > 0$, $y = x$ with a slope of $1 > 0$, which is an increasing function;)
 - 当 $x < 0$ 时, $y = -x$, 斜率为 $-1 < 0$, 是减函数;(When $x < 0$, $y = -x$ with a slope of $-1 < 0$, which is a decreasing function;)
 - 当 $x \in \mathbb{R}$ 时, 函数在 $x = 0$ 处转折, 整体非增非减。(When $x \in \mathbb{R}$, the function has a turning point at $x = 0$ and is neither increasing nor decreasing overall.)
- 故 B 正确。(Thus, B is correct.)

第 12 题

题目(Question): 有理不等式 $\frac{2x+1}{x-2} \leq 0$ 的解集是? (What is the solution set of the rational inequality $\frac{2x+1}{x-2} \leq 0$?)

- A $[-\frac{1}{2}, 2)$
- B $(-\infty, -\frac{1}{2}] \cup (2, +\infty)$
- C $[-\frac{1}{2}, 2]$
- D $(-\infty, -\frac{1}{2}]$

答案(Answer): A

解析(Explanation):

- 有理不等式求解步骤:(Steps to solve rational inequalities;)
 1. 分母不为 0: $x - 2 \neq 0$, 即 $x \neq 2$; (Ensure the denominator is not zero: $x - 2 \neq 0$, i.e., $x \neq 2$;)
 2. 转化为整式不等式: $(2x + 1)(x - 2) \leq 0$ (分子分母异号时分数 ≤ 0); (Convert to



an integral inequality: $(2x+1)(x-2) \leq 0$ (the fraction ≤ 0 when the numerator and denominator have opposite signs);

3. 求解整式不等式:因式零点为 $x = -\frac{1}{2}$ 和 $x = 2$, 解为 $-\frac{1}{2} \leq x \leq 2$; (Solve the integral inequality: the zeros of the factors are $x = -\frac{1}{2}$ and $x = 2$, and the solution is $-\frac{1}{2} \leq x \leq 2$.)

4. 结合分母不为 0, 最终解集为 $[-\frac{1}{2}, 2)$ 。(Combined with $x \neq 2$, the final solution set is $[-\frac{1}{2}, 2)$.)

- 故 A 正确。(Thus, A is correct.)

第 13 题

题目(Question):已知 $a = 2 - \sqrt{3}$, $b = 2 + \sqrt{3}$, 则 a 与 b 的算术平均数(arithmetic mean)是? (Given $a = 2 - \sqrt{3}$ and $b = 2 + \sqrt{3}$, what is the arithmetic mean of a and b?)

A:1

B:2

C: ± 1

D:-1

答案(Answer):B

解析(Explanation):

- 算术平均数定义:两数 a、b 的算术平均数为 $\frac{a+b}{2}$ 。

(Definition of arithmetic mean: The arithmetic mean of two numbers a and b is $\frac{a+b}{2}$.)

- 代入计算:(Substitute and calculate:)

$$\frac{a+b}{2} = \frac{(2-\sqrt{3}) + (2+\sqrt{3})}{2} = \frac{4}{2} = 2$$

- 故 B 正确。(Thus, B is correct.)

第 14 题

题目(Question):若直线 l 经过点 $A(-2,3)$ 和 $B(3,1)$, 则直线 l 的斜率(slope)是?

(If straight line l passes through points $A(-2,3)$ and $B(3,1)$, what is the slope of line l ?)

A $\frac{2}{5}$

B $\frac{5}{2}$

C $-\frac{2}{5}$

D $-\frac{5}{2}$

答案(Answer):C

解析(Explanation):

- 过两点 (x_1, y_1) 和 (x_2, y_2) 的直线斜率公式: $k = \frac{y_2 - y_1}{x_2 - x_1}$ 。

(Slope formula of a straight line passing through two points (x_1, y_1) and (x_2, y_2) : $k =$

$$\frac{y_2 - y_1}{x_2 - x_1} .)$$

- 代入 $A(-2,3)$ ($x_1 = -2$, $y_1 = 3$)和 $B(3,1)$ ($x_2 = 3$, $y_2 = 1$):

(Substitute $A(-2,3)$ ($x_1 = -2$, $y_1 = 3$) and $B(3,1)$ ($x_2 = 3$, $y_2 = 1$):

$$k = \frac{1 - 3}{3 - (-2)} = \frac{-2}{5} = -\frac{2}{5}$$

- 故 C 正确。(Thus, C is correct.)

第 15 题

题目(Question):若 α 是 x 轴与过点 $P(1,3)$ 的直线的夹角(angle), 则 $\sin \alpha = ?$ (If α is the

angle between the x -axis and the straight line passing through point $P(1,3)$, then $\sin \alpha = ?$)

A 3

B $\frac{3\sqrt{10}}{10}$

C $\frac{1}{3}$

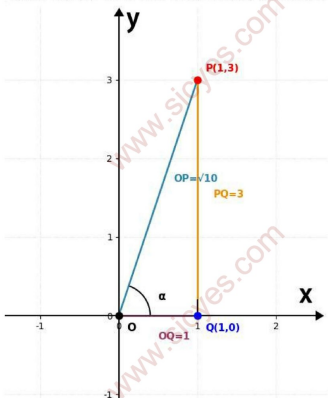


$$D \frac{\sqrt{10}}{10}$$

答案(Answer):B

解析(Explanation):

Right Triangle OPQ: Side Lengths and Angle Label



- 过点 $P(1,3)$ 作 x 轴垂线, 垂足为 $Q(1,0)$, 直角三角形 OPQ 中:(Draw a perpendicular from $P(1,3)$ to the x -axis, with the foot of the perpendicular being $Q(1,0)$. In the right triangle OPQ :)
 ○ 对边: $PQ = 3$; (Opposite side: $PQ = 3$;)
 ○ 邻边: $OQ = 1$; (Adjacent side: $OQ = 1$;)
 ○ 斜边: $OP = \sqrt{1^2 + 3^2} = \sqrt{10}$ 。 (Hypotenuse: $OP = \sqrt{1^2 + 3^2} = \sqrt{10}$.)
- 正弦函数定义: $\sin \alpha = \frac{\text{对边}}{\text{斜边}} = \frac{3}{\sqrt{10}} = \frac{3\sqrt{10}}{10}$, 故 B 正确。(According to the definition of sine function: $\sin \alpha = \frac{\text{opposite side}}{\text{hypotenuse}} = \frac{3}{\sqrt{10}} = \frac{3\sqrt{10}}{10}$, so B is correct.)

第 16 题

题目(Question): 下列四个数列中, 等差数列(arithmetic sequence)的个数是? (Among the following four sequences, how many are arithmetic sequences?)

(1) 7, 13, 19, 25 (2) 2, 4, 7, 11 (3) -1, -3, -5, -7 (4) 2, 4, 8, 16

A 2

B 3

C 1

D 4

答案(Answer): A

解析(Explanation):

- 等差数列定义: 从第二项起, 每一项与前一項的差为常数(公差 d)。(Definition of arithmetic sequence: From the second term onwards, the difference between each term and its previous term is a constant (common difference d).)
- 逐个分析: (Analyze one by one:)
 1. 数列 7,13,19,25: 差值均为 6 ($d=6$), 是等差数列; (Sequence 7,13,19,25: differences are all 6 ($d=6$), which is an arithmetic sequence;)
 2. 数列 2,4,7,11: 差值为 2、3、4(不相等), 不是等差数列; (Sequence 2,4,7,11: differences are 2, 3, 4 (not equal), which is not an arithmetic sequence;)
 3. 数列 -1,-3,-5,-7: 差值均为 -2 ($d=-2$), 是等差数列; (Sequence -1,-3,-5,-7: differences are all -2 ($d=-2$), which is an arithmetic sequence;)
 4. 数列 2,4,8,16: 差值为 2、4、8(不相等), 是等比数列, 不是等差数列。 (Sequence 2,4,8,16: differences are 2, 4, 8 (not equal), which is a geometric sequence, not an



arithmetic sequence.)

- 综上，等差数列有 2 个，A 正确。(In summary, there are 2 arithmetic sequences, and A is correct.)

第 17 题

题目(Question):平面上点 $A(3, -2)$ 到点 $B(-5, -1)$ 的距离(distance)是 $|AB| = ?$ (What is the distance $|AB|$ between point $A(3, -2)$ and point $B(-5, -1)$ on the plane?)

A 65

B $\sqrt{65}$

C 13

D $\sqrt{13}$

答案(Answer):B

解析(Explanation):

- 平面两点间距离公式: $|AB| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ 。(Distance formula between two points on the plane: $|AB| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.)
- 代入 $A(3, -2)$ ($x_1 = 3$, $y_1 = -2$) 和 $B(-5, -1)$ ($x_2 = -5$, $y_2 = -1$): (Substitute $A(3, -2)$ ($x_1 = 3$, $y_1 = -2$) and $B(-5, -1)$ ($x_2 = -5$, $y_2 = -1$):)

$$|AB| = \sqrt{(-5 - 3)^2 + (-1 - (-2))^2} = \sqrt{(-8)^2 + 1^2} = \sqrt{64 + 1} = \sqrt{65}$$

- 故 B 正确。(Thus, B is correct.)

第 18 题

题目(Question):若直线 l 的倾斜角(angle of inclination)为 45° ，且过点 $(0, 2)$ ，则直线 l 的方程(equation)是? (If the angle of inclination of straight line l is 45° and it passes through point $(0,$

2), what is the equation of line l?)

A $y = 2x - 2$

B $y = x + 2$

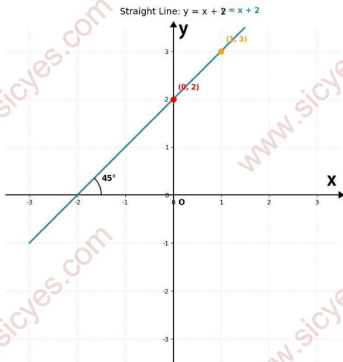
C $y = 2x + 2$

D $y = x - 2$

答案(Answer):B

解析(Explanation):

- 直线倾斜角与斜率关系: $k = \tan\theta$ (θ 为倾斜角)。(Relationship between the angle of inclination and slope: $k = \tan\theta$ (θ is the angle of inclination).)
- 倾斜角 $\theta = 45^\circ$, 故斜率 $k = \tan 45^\circ = 1$ 。(Given $\theta = 45^\circ$, the slope $k = \tan 45^\circ = 1$.)
- 直线过点(0,2)(y轴截距 $b = 2$), 代入斜截式 $y = kx + b$ 得: $y = x + 2$, B 正确。(The line passes through (0, 2) (y-intercept $b = 2$), substitute into the slope-intercept form $y = kx + b$ to get $y = x + 2$, so B is correct.)



第 19 题

题目(Question):若圆的方程为 $x^2 + y^2 - 4x - 3 = 0$, 则该圆的圆心(center)和半径(radius)分别是? (If the equation of a circle is $x^2 + y^2 - 4x - 3 = 0$, what are the center and radius of the circle?)

A (0, 2) 和 $\sqrt{7}$



B (0, 2) 和 7

C (2, 0) 和 $\sqrt{7}$

D (2, 0) 和 7

答案(Answer):C

解析(Explanation):

- 圆的一般方程化为标准方程(配方法): (Convert the general equation to the standard equation by completing the square.)

$$(x^2 - 4x + 4) + y^2 = 3 + 4 \Rightarrow (x - 2)^2 + y^2 = 7$$

- 圆的标准方程为 $(x - h)^2 + (y - k)^2 = r^2$, 其中 (h, k) 为圆心, r 为半径。(The standard equation of a circle is $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center and r is the radius.)
- 对比得: 圆心(2, 0), 半径 $r = \sqrt{7}$, 故 C 正确。(By comparison: the center is (2, 0) and the radius $r = \sqrt{7}$, so C is correct.)

第 20 题

题目(Question): 已知 $\sin \alpha = \frac{4}{5}$, 且 α 是第二象限角(second quadrant angle), 则下列说法正确的是? (Given $\sin \alpha = \frac{4}{5}$ and α is a second quadrant angle, which of the following statements is correct?)

A $\cos \alpha = \frac{3}{5}$

B $\cos \alpha = -\frac{3}{5}$

C $\tan \alpha = -\frac{3}{4}$

D $\tan \alpha = \frac{4}{3}$

答案(Answer):B

解析(Explanation):

- 同角三角函数基本关系: $\sin^2\alpha + \cos^2\alpha = 1$ 。(Basic relation of trigonometric functions of the same angle: $\sin^2\alpha + \cos^2\alpha = 1$.)

- 计算 $\cos\alpha$:(Calculate $\cos\alpha$:)

$$\cos\alpha = \pm\sqrt{1 - \sin^2\alpha} = \pm\sqrt{1 - \left(\frac{4}{5}\right)^2} = \pm\frac{3}{5}$$

- α 是第二象限角, 余弦值为负, 故 $\cos\alpha = -\frac{3}{5}$, B 正确。(Since α is a second quadrant angle, the cosine value is negative, so $\cos\alpha = -\frac{3}{5}$, and B is correct.)
- 进一步计算 $\tan\alpha = \frac{\sin\alpha}{\cos\alpha} = -\frac{4}{3}$, 故 C、D 错误。(Further calculate $\tan\alpha = \frac{\sin\alpha}{\cos\alpha} = -\frac{4}{3}$, so C and D are incorrect.)

第 21 题

题目(Question):在等差数列 $\{a_n\}$ 中, 若 $a_2 = 1$, $a_4 = 5$, 则首项 a_1 和公差 d 分别是? (In the arithmetic sequence $\{a_n\}$, if $a_2 = 1$ and $a_4 = 5$, what are the first term a_1 and common difference d respectively?)

A -1, 2

B 1, -2

C -1, -2

D 1, 2

答案(Answer):A

解析(Explanation):

- 等差数列通项公式: $a_n = a_1 + (n - 1)d$ 。(General term formula of arithmetic sequence: $a_n = a_1 + (n - 1)d$.)
- 列方程组:(Set up a system of equations:)



$$\begin{cases} a_1 + d = 1 \\ a_1 + 3d = 5 \end{cases}$$

- 两式相减得 $2d = 4$, 解得 $d = 2$; 代入 $a_1 + d = 1$ 得 $a_1 = -1$ 。(Subtract the first equation from the second to get $2d = 4$, so $d = 2$; substitute into $a_1 + d = 1$ to get $a_1 = -1$.)
- 故首项 $a_1 = -1$ 、公差 $d = 2$, A 正确。(Thus, the first term $a_1 = -1$ and common difference $d = 2$, so A is correct.)

第 22 题

题目(Question): 设 a 、 b 、 c 为三个实数(real numbers), 则下列说法正确的是? (Let a , b , c be three real numbers. Then which of the following statements is correct?)

- A 若 $a > b$, 则 $ac > bc$ (If $a > b$, then $ac > bc$)
- B 若 $a > b$, 则 $a + c > b + c$ (If $a > b$, then $a + c > b + c$)
- C 若 $ac > bc$, 则 $a > b$ (If $ac > bc$, then $a > b$)
- D 若 $a > b$, 则 $a^2 > b^2$ (If $a > b$, then $a^2 > b^2$)

答案(Answer):B

解析(Explanation):

- 不等式基本性质:(Basic properties of inequalities:)
- 选项 A: 当 $c = 0$ 时 $ac = bc$, 当 $c < 0$ 时不等号反向, 故 A 错误; (Option A: When $c = 0$, $ac = bc$; when $c < 0$, the inequality sign reverses, so A is incorrect.)
- 选项 B: 不等式两边加同一数, 不等号方向不变, 故 $a + c > b + c$ 成立, B 正确; (Option B: Adding the same number to both sides of an inequality, the sign remains unchanged, so $a + c > b + c$ holds, and B is correct.)
- 选项 C: 当 $c < 0$ 时, 两边除以 c 不等号反向, 得 $a < b$, 故 C 错误; (Option C: When

$c < 0$, dividing both sides by c reverses the sign, resulting in $a < b$, so C is incorrect;)

- 选项 D:反例: $a = 1$ 、 $b = -2$, $a > b$ 但 $a^2 < b^2$, 故 D 错误。(Option D:

Counterexample: $a = 1$, $b = -2$, $a > b$ but $a^2 < b^2$, so D is incorrect.)

第 23 题

题目(Question):关于指数函数(exponential function) $y = a^x$ ($a > 0$ 且 $a \neq 1$), 下列说法不正确的是? (About the exponential function $y = a^x$ ($a > 0$ and $a \neq 1$), which of the following statements is incorrect?)

- A 值域(range)是 $(0, +\infty)$ (The range is $(0, +\infty)$)
- B 图像过点(passes through the point)(0, 1)(The graph passes through the point (0, 1))
- C 在定义域(domain)上是增函数(is an increasing function over the domain)
- D 定义域(domain)是 $\mathbb{R}$ (The domain is $\mathbb{R}$)

答案(Answer):C

解析(Explanation):

- 指数函数性质:(Properties of exponential functions:
 - 定义域为 $\mathbb{R}$, 故 D 正确; (Domain is $\mathbb{R}$, so D is correct;)
 - 值域为 $(0, +\infty)$, 故 A 正确; (Range is $(0, +\infty)$, so A is correct;)
 - 当 $x = 0$ 时, $y = a^0 = 1$, 图像过(0, 1), 故 B 正确; (When $x = 0$, $y = a^0 = 1$, the graph passes through (0, 1), so B is correct;)
 - 单调性: $a > 1$ 时为增函数, $0 < a < 1$ 时为减函数, 故 C 错误。(Monotonicity: Increasing when $a > 1$, decreasing when $0 < a < 1$, so C is incorrect.)

第 24 题

题目(Question):若 $\sin \alpha = \frac{3}{5}$, 且 α 是第一象限角(first quadrant angle), 则 $\sin 2\alpha = ?$ (If



$\sin \alpha = \frac{3}{5}$ and α is a first quadrant angle, then $\sin 2\alpha = ?$

A $\frac{12}{25}$

B $\frac{24}{25}$

C $\frac{18}{25}$

D $\frac{7}{25}$

答案(Answer):B

解析(Explanation):

- 二倍角公式: $\sin 2\alpha = 2\sin\alpha\cos\alpha$ 。 (Double-angle formula: $\sin 2\alpha = 2\sin\alpha\cos\alpha$.)
- 求 $\cos\alpha$: α 是第一象限角, $\cos\alpha = \sqrt{1 - \sin^2\alpha} = \sqrt{1 - \left(\frac{3}{5}\right)^2} = \frac{4}{5}$ 。 (Find $\cos\alpha$: α is a first quadrant angle, so $\cos\alpha = \sqrt{1 - \sin^2\alpha} = \frac{4}{5}$.)
- 代入公式: $\sin 2\alpha = 2 \times \frac{3}{5} \times \frac{4}{5} = \frac{24}{25}$, B 正确。 (Substitute into the formula: $\sin 2\alpha = 2 \times \frac{3}{5} \times \frac{4}{5} = \frac{24}{25}$, so B is correct.)

第 25 题

题目(Question):直线 $l_1: 3x - y + 8 = 0$ 与直线 $l_2: x + 2y - 9 = 0$ 的交点(intersection point)

坐标是? (What are the coordinates of the intersection point of line $l_1: 3x - y + 8 = 0$ and line $l_2: x + 2y - 9 = 0$?)

A (-5, 1)

B (1, -5)

C (5, -1)

D (-1, 5)

答案(Answer):D

解析(Explanation):

- 解方程组求交点:(Solve the system of equations to find the intersection:)

$$\begin{cases} 3x - y = -8 \\ x + 2y = 9 \end{cases}$$

- 由第一式得 $y = 3x + 8$, 代入第二式:(From the first equation, $y = 3x + 8$, substitute into the second equation:)

$$x + 2(3x + 8) = 9 \Rightarrow 7x = -7 \Rightarrow x = -1$$

- 代入 $y = 3x + 8$ 得 $y = 5$, 故交点坐标为 $(-1, 5)$, D 正确。(Substitute into $y = 3x + 8$ to get $y = 5$, so the intersection coordinates are $(-1, 5)$, and D is correct.)

第 26 题

题目(Question):已知双曲线(hyperbola)的方程为 $\frac{x^2}{64} - \frac{y^2}{16} = 1$, 则下列说法正确的是? (Given the equation of the hyperbola is $\frac{x^2}{64} - \frac{y^2}{16} = 1$, which of the following statements is correct?)

- A 焦距(focal distance)是 $8\sqrt{5}$ (The focal distance is $8\sqrt{5}$)
- B 实半轴长(length of semi-major axis)是 4(The length of the semi-major axis is 4)
- C 离心率(eccentricity)是 $\sqrt{5}$ (The eccentricity is $\sqrt{5}$)
- D 虚半轴长(length of semi-minor axis)是 8(The length of the semi-minor axis is 8)

答案(Answer):A

解析(Explanation):

- 双曲线标准方程 $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (焦点在 x 轴):(Standard equation of hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (foci on the x-axis):)
 - $a^2 = 64$, 实半轴长 $a = 8$, 故 B 错误; ($a^2 = 64$, length of semi-major axis $a = 8$, so B is incorrect;)
 - $b^2 = 16$, 虚半轴长 $b = 4$, 故 D 错误; ($b^2 = 16$, length of semi-minor axis $b = 4$, so D is incorrect;)
 - 焦距相关: $c^2 = a^2 + b^2 = 64 + 16 = 80$, $c = 4\sqrt{5}$, 焦距 $2c = 8\sqrt{5}$, 故 A 正确;



(Focal distance: $c^2 = a^2 + b^2 = 80$, $c = 4\sqrt{5}$, focal distance $2c = 8\sqrt{5}$, so A is correct.)

- 离心率 $e = \frac{c}{a} = \frac{\sqrt{5}}{2}$, 故 C 错误。(Eccentricity $e = \frac{c}{a} = \frac{\sqrt{5}}{2}$, so C is incorrect.)

第 27 题

题目(Question): 下列不等式正确的是? (Which of the following inequalities is correct?)

A $2.1^{-2} > 1.2^{-2}$

B $0.75^{-0.2} > 0.75^{-0.4}$

C $3^{2.25} > 3^3$

D $2.1^{\frac{2}{3}} > 1.2^{\frac{2}{3}}$ ($2.1^{\frac{2}{3}} > 1.2^{\frac{2}{3}}$)

答案(Answer): D

解析(Explanation):

- 利用函数单调性分析: (Analyze using function monotonicity):
 - 选项 A: 幂函数 $y = x^{-2}$ 在 $(0, +\infty)$ 上减, $2.1 > 1.2$, 故 $2.1^{-2} < 1.2^{-2}$, A 错误;
(Option A: Power function $y = x^{-2}$ is decreasing on $(0, +\infty)$, $2.1 > 1.2$, so $2.1^{-2} < 1.2^{-2}$, A is incorrect.)
 - 选项 B: 指数函数 $y = 0.75^x$ 减, $-0.2 > -0.4$, 故 $0.75^{-0.2} < 0.75^{-0.4}$, B 错误;
(Option B: Exponential function $y = 0.75^x$ is decreasing, $-0.2 > -0.4$, so $0.75^{-0.2} < 0.75^{-0.4}$, B is incorrect.)
 - 选项 C: 指数函数 $y = 3^x$ 增, $2.25 < 3$, 故 $3^{2.25} < 3^3$, C 错误; (Option C: Exponential function $y = 3^x$ is increasing, $2.25 < 3$, so $3^{2.25} < 3^3$, C is incorrect.)
 - 选项 D: 幂函数 $y = x^{\frac{2}{3}}$ 在 $(0, +\infty)$ 上增, $2.1 > 1.2$, 故 $2.1^{\frac{2}{3}} > 1.2^{\frac{2}{3}}$, D 正确。
(Option D: Power function $y = x^{\frac{2}{3}}$ is increasing on $(0, +\infty)$, $2.1 > 1.2$, so $2.1^{\frac{2}{3}} > 1.2^{\frac{2}{3}}$.)

$1.2^{\frac{2}{3}}$, D is correct.)

第 28 题

题目(Question): 圆心在 $(-3, 2)$, 半径为 4 的圆的方程是? (What is the equation of the circle with center at $(-3, 2)$ and radius 4?)

A $(x + 3)^2 + (y - 2)^2 = 4$

B $(x - 3)^2 + (y + 2)^2 = 4$

C $(x - 3)^2 + (y + 2)^2 = 16$

D $(x + 3)^2 + (y - 2)^2 = 16$

答案(Answer): D

解析(Explanation):

- 圆的标准方程: $(x - h)^2 + (y - k)^2 = r^2$, 其中 (h, k) 为圆心, r 为半径。(Standard equation of a circle: $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center and r is the radius.)

- 代入圆心 $(-3, 2)$ 、半径 4: (Substitute center $(-3, 2)$ and radius 4:)

$$(x - (-3))^2 + (y - 2)^2 = 4^2 \Rightarrow (x + 3)^2 + (y - 2)^2 = 16$$

- 故 D 正确。(Thus, D is correct.)

第 29 题

题目(Question): 设 $a > 0$, $b > 0$ 为实数, 则下列说法不正确的是? (Let $a > 0$ and $b > 0$ be real numbers. Then which of the following statements is incorrect?)

A $\ln(a^b) = b \ln a$

B $\log_a b = \frac{\ln a}{\ln b}$ ($a \neq 1, b \neq 1$)

C $\ln\left(\frac{a}{b}\right) = \ln a - \ln b$



D $\ln(ab) = \ln a + \ln b$

答案(Answer):B

解析(Explanation):

- 对数运算性质:(Operation properties of logarithms:)
 - 选项 A: 幂的对数: $\ln(a^b) = b \ln a$, 正确; (Option A: Logarithm of a power: $\ln(a^b) = b \ln a$, correct;)
 - 选项 B: 换底公式: $\log_a b = \frac{\ln b}{\ln a}$ (非 $\frac{\ln a}{\ln b}$), 颠倒分子分母, 错误; (Option B: Change of base formula: $\log_a b = \frac{\ln b}{\ln a}$ (not $\frac{\ln a}{\ln b}$), reverses numerator and denominator, incorrect;)
 - 选项 C: 商的对数: $\ln\left(\frac{a}{b}\right) = \ln a - \ln b$, 正确; (Option C: Logarithm of a quotient: $\ln\left(\frac{a}{b}\right) = \ln a - \ln b$, correct;)
 - 选项 D: 积的对数: $\ln(ab) = \ln a + \ln b$, 正确。 (Option D: Logarithm of a product: $\ln(ab) = \ln a + \ln b$, correct.)

第 30 题

题目(Question): 已知正弦函数(sine function) $y = \sin x$, 下列说法不正确的是? (Given the sine function $y = \sin x$, which of the following statements is incorrect?)

- A 最大值和最小值分别为 1 和 -1(The maximum and minimum values are 1 and -1 respectively)
 B 该函数是周期函数(The function is a periodic function) C 该函数是偶函数(The function is even)
 D 定义域为 $\mathbb{R}$ (The domain is $\mathbb{R}$)

答案(Answer):C

解析(Explanation):

- 正弦函数核心性质:(Core properties of the sine function $y = \sin x$:)
 - 值域 $[-1, 1]$, 最大值 1、最小值 -1, A 正确; (Range $[-1, 1]$, maximum 1, minimum

-1, A correct;)

- 最小正周期 2π , 是周期函数, B 正确; (Minimum positive period 2π , it is a periodic function, B correct;)
- 奇偶性: $\sin(-x) = -\sin x$, 是奇函数, 非偶函数, C 错误; (Parity: $\sin(-x) = -\sin x$, it is an odd function, not even, C incorrect;)
- 定义域为 $\mathbb{R}$, D 正确。 (Domain is $\mathbb{R}$, D correct.)

第 31 题

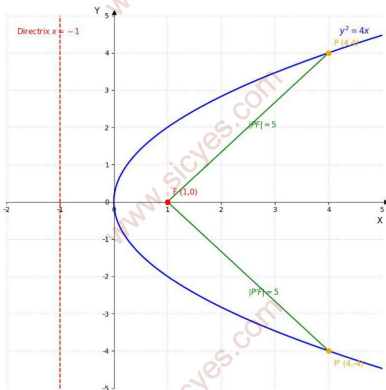
题目(Question): 设抛物线(parabola) $y^2 = 4x$ 的焦点为 F , 点 P 在抛物线上, 且 P 的横坐标(horizontal coordinate)为 4, 则 $|PF| = ?$ (Let F be the focus of the parabola $y^2 = 4x$.

Suppose point P is on the parabola and the horizontal coordinate of P is 4. Then $|PF| = ?$)

A 2 B 3 C 4 D 5

答案(Answer):D

解析(Explanation):





- 抛物线 $y^2 = 2px$ (焦点在 x 轴) 的焦点坐标: $F\left(\frac{p}{2}, 0\right)$ 。(Focus coordinates of parabola $y^2 = 2px$ (foci on the x -axis): $F\left(\frac{p}{2}, 0\right)$.)
- 已知 $y^2 = 4x$, 则 $2p = 4$ ($p = 2$), 焦点 $F(1, 0)$ 。(Given $y^2 = 4x$, $2p = 4$ ($p = 2$), focus $F(1, 0)$.)
- 点 P 横坐标为 4, 代入抛物线方程得 $y^2 = 16$, 故 $P(4, \pm 4)$ 。(The horizontal coordinate of P is 4, substitute into the parabola equation to get $y^2 = 16$, so $P(4, \pm 4)$.)
- 距离公式计算 $|PF|$:(Calculate $|PF|$ using the distance formula:)

$$|PF| = \sqrt{(4-1)^2 + (\pm 4-0)^2} = \sqrt{9+16} = 5$$

- 故 D 正确。(Thus, D is correct.)

第 32 题

题目(Question): 已知两条直线 $l_1: ax + (a-1)y + 3 = 0$ 和 $l_2: 2x + ay - 1 = 0$, 若 $l_1 \perp l_2$ (垂直, perpendicular), 则 a 的值为? (Given two straight lines $l_1: ax + (a-1)y + 3 = 0$ and $l_2: 2x + ay - 1 = 0$. If $l_1 \perp l_2$ (perpendicular), then the value of a is?)

A -1 B -1 或 1 (-1 or 1) C 1 D 0 或 -1 (0 or -1)

答案(Answer): D

解析(Explanation):

- 两直线垂直条件: 对于 $l_1: A_1x + B_1y + C_1 = 0$ 和 $l_2: A_2x + B_2y + C_2 = 0$, 垂直充要条件为 $A_1A_2 + B_1B_2 = 0$ 。(Condition for two lines to be perpendicular: For $l_1: A_1x + B_1y + C_1 = 0$ and $l_2: A_2x + B_2y + C_2 = 0$, the necessary and sufficient condition is $A_1A_2 + B_1B_2 = 0$.)
- 代入系数:(Substitute the coefficients:)

$$a \times 2 + (a - 1) \times a = 0 \Rightarrow a^2 + a = 0 \Rightarrow a(a + 1) = 0$$

- 解得 $a = 0$ 或 $a = -1$, 故 D 正确。(Solve to get $a = 0$ or $a = -1$, so D is correct.)

第 33 题

题目(Question): 点 $P(-1,2)$ 和 $Q(3,1)$ 之间的距离(distance)是? (What is the distance between points $P(-1,2)$ and $Q(3,1)$?)

A $\sqrt{5}$ ($\sqrt{5}$) B $\sqrt{17}$ ($\sqrt{17}$) C $\sqrt{13}$ ($\sqrt{13}$) D 5

答案(Answer):B

解析(Explanation):

- 平面两点间距离公式: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ 。 (Distance formula between two points on the plane: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.)

- 代入 $P(-1,2)$ ($x_1 = -1$, $y_1 = 2$)和 $Q(3,1)$ ($x_2 = 3$, $y_2 = 1$);(Substitute $P(-1,2)$ ($x_1 = -1$, $y_1 = 2$) and $Q(3,1)$ ($x_2 = 3$, $y_2 = 1$);)

$$d = \sqrt{(3 - (-1))^2 + (1 - 2)^2} = \sqrt{4^2 + (-1)^2} = \sqrt{16 + 1} = \sqrt{17}$$

- 故 B 正确。(Thus, B is correct.)

第 34 题

题目(Question): 设 $\cos \alpha = -\frac{\sqrt{5}}{5}$, 且 $\alpha \in \left(\frac{\pi}{2}, \pi\right)$, 则 $\sin\left(\frac{\pi}{6} + \alpha\right) = ?$ (Let $\cos \alpha = -\frac{\sqrt{5}}{5}$ and $\alpha \in \left(\frac{\pi}{2}, \pi\right)$. Then $\sin\left(\frac{\pi}{6} + \alpha\right) = ?$)

A $\frac{2\sqrt{5}-\sqrt{15}}{10}$

B $\frac{2\sqrt{15}+\sqrt{5}}{10}$

C $\frac{2\sqrt{15}-\sqrt{5}}{10}$

D $\frac{2\sqrt{5}+\sqrt{15}}{10}$

答案(Answer):C

解析(Explanation):

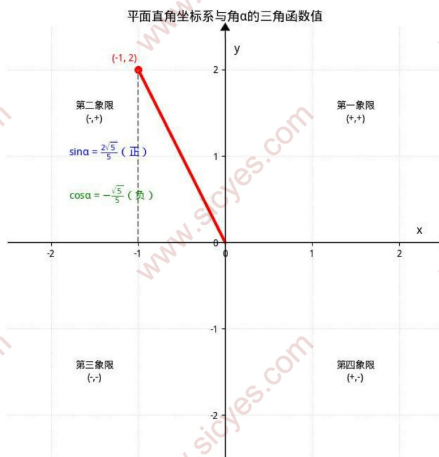


- 步骤 1: 求 $\sin \alpha$ (Step 1: Find $\sin \alpha$):

○ 因 $\alpha \in \left(\frac{\pi}{2}, \pi\right)$ (第二象限), $\sin \alpha > 0$ (第二象限正弦值为正)。(Since $\alpha \in \left(\frac{\pi}{2}, \pi\right)$ (second quadrant), $\sin \alpha > 0$ (sine value is positive in the second quadrant).)

○ 由同角三角函数关系 $\sin^2 \alpha + \cos^2 \alpha = 1$: (Using the relation of trigonometric functions of the same angle $\sin^2 \alpha + \cos^2 \alpha = 1$:)

$$\sin \alpha = \sqrt{1 - \cos^2 \alpha} = \sqrt{1 - \left(-\frac{\sqrt{5}}{5}\right)^2} = \sqrt{1 - \frac{1}{5}} = \frac{2\sqrt{5}}{5}$$



- 步骤 2: 应用两角和的正弦公式 (Step 2: Apply the sine addition formula):

○ 两角和公式: $\sin(A + B) = \sin A \cos B + \cos A \sin B$ 。(Sine addition formula: $\sin(A + B) = \sin A \cos B + \cos A \sin B$.)

○ 代入 $A = \frac{\pi}{6}$, $B = \alpha$: (Substitute $A = \frac{\pi}{6}$ and $B = \alpha$:)

$$\sin\left(\frac{\pi}{6} + \alpha\right) = \sin\frac{\pi}{6}\cos\alpha + \cos\frac{\pi}{6}\sin\alpha$$

$$= \frac{1}{2} \times \left(-\frac{\sqrt{5}}{5}\right) + \frac{\sqrt{3}}{2} \times \frac{2\sqrt{5}}{5}$$

$$= -\frac{\sqrt{5}}{10} + \frac{2\sqrt{15}}{10} = \frac{2\sqrt{15}-\sqrt{5}}{10}$$

- 故 C 正确。(Thus, C is correct.)

第 35 题

题目(Question): 下列说法正确的是? (Which of the following statements is correct?)

A $y = (\sqrt{x})^2$ 和 $y = \sqrt{x^2}$ 是同一函数($y = (\sqrt{x})^2$ and $y = \sqrt{x^2}$ are the same function)

B $y = \frac{x^4-1}{x^2+1}$ 和 $y = x^2 - 1$ 是同一函数($y = \frac{x^4-1}{x^2+1}$ and $y = x^2 - 1$ are the same function)

C $y = 1$ 和 $y = x^2$ 是同一函数($y = 1$ and $y = x^2$ are the same function)

D $y = x$ 和 $y = \sqrt{x^2}$ 是同一函数($y = x$ and $y = \sqrt{x^2}$ are the same function)

答案(Answer):B

解析(Explanation):

- 同一函数的定义: 定义域和对应关系完全相同, 与变量名称无关。(Definition of the same function: The domain and corresponding relation are exactly the same, regardless of the variable name.)
- 逐个分析:(Analyze one by one:)
 - 选项 A: $y = (\sqrt{x})^2$ 定义域为 $x \geq 0$, 对应关系为 $y = x$; $y = \sqrt{x^2} = |x|$ 定义域为 $\mathbb{R}$, 对应关系为 $y = |x|$, 定义域和对应关系均不同, 不是同一函数; (Option A: The domain of $y = (\sqrt{x})^2$ is $x \geq 0$ and the corresponding relation is $y = x$; the domain of $y = \sqrt{x^2} = |x|$ is $\mathbb{R}$ and the corresponding relation is $y = |x|$. The domain and corresponding relation are different, not the same function;)
 - 选项 B: $y = \frac{x^4-1}{x^2+1} = \frac{(x^2-1)(x^2+1)}{x^2+1} = x^2 - 1$ ($x^2 + 1 \neq 0$, 定义域为 $\mathbb{R}$); $y = x^2 - 1$



定义域为 $\mathbb{R}$ ，对应关系相同，是同一函数；(Option B: $y = \frac{x^4-1}{x^2+1} = \frac{(x^2-1)(x^2+1)}{x^2+1} = x^2 - 1$

1 ($x^2 + 1 \neq 0$ ，domain is $\mathbb{R}$); the domain of $y = x^2 - 1$ is $\mathbb{R}$ ，and the corresponding relation is the same, so they are the same function;)

- 选项 C: $y = 1$ 对应关系为“常值 1”， $y = x^2$ 对应关系为“平方运算”，对应关系不同，不是同一函数；(Option C: The corresponding relation of $y = 1$ is “constant value 1”，and that of $y = x^2$ is “square operation”. The corresponding relations are different, not the same function;)
- 选项 D: $y = x$ 对应关系为 $y = x$ ， $y = \sqrt{x^2} = |x|$ 对应关系为 $y = |x|$ ，对应关系不同，不是同一函数。(Option D: The corresponding relation of $y = x$ is $y = x$ ，and that of $y = \sqrt{x^2} = |x|$ is $y = |x|$. The corresponding relations are different, not the same function.)

第 36 题

题目(Question): 设 $\sin \alpha = -\frac{12}{13}$ ，且 $\alpha \in (\pi, \frac{3\pi}{2})$ ，则 $\cos \frac{\alpha}{2} = ?$ (Let $\sin \alpha = -\frac{12}{13}$ and $\alpha \in (\pi, \frac{3\pi}{2})$. Then $\cos \frac{\alpha}{2} = ?$)

A $\frac{3\sqrt{13}}{13}$

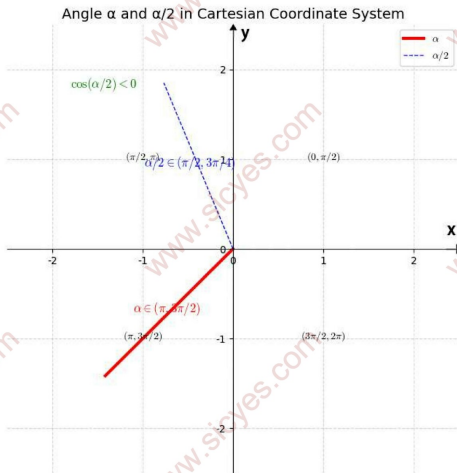
B $-\frac{3\sqrt{13}}{13}$

C $-\frac{2\sqrt{13}}{13}$

D $\frac{2\sqrt{13}}{13}$

答案(Answer): C

解析(Explanation):



- 步骤 1: 确定 $\frac{\alpha}{2}$ 的象限 (Step 1: Determine the quadrant of $\frac{\alpha}{2}$):
 - 因 $\alpha \in (\pi, \frac{3\pi}{2})$, 两边除以 2 得 $\frac{\alpha}{2} \in (\frac{\pi}{2}, \frac{3\pi}{4})$ (第二象限), 第二象限中 $\cos \frac{\alpha}{2} < 0$ 。(Since $\alpha \in (\pi, \frac{3\pi}{2})$, divide both sides by 2 to get $\frac{\alpha}{2} \in (\frac{\pi}{2}, \frac{3\pi}{4})$ (second quadrant), where $\cos \frac{\alpha}{2} < 0$.)
- 步骤 2: 求 $\cos \alpha$ (Step 2: Find $\cos \alpha$):
 - $\alpha \in (\pi, \frac{3\pi}{2})$ (第三象限), $\cos \alpha < 0$, 由 $\sin^2 \alpha + \cos^2 \alpha = 1$: ($\alpha \in (\pi, \frac{3\pi}{2})$ (third quadrant), $\cos \alpha < 0$, using $\sin^2 \alpha + \cos^2 \alpha = 1$:)
$$\cos \alpha = -\sqrt{1 - \sin^2 \alpha} = -\sqrt{1 - \left(-\frac{12}{13}\right)^2} = -\sqrt{1 - \frac{144}{169}} = -\frac{5}{13}$$
- 步骤 3: 应用半角公式 (Step 3: Apply the half-angle formula):



- 半角公式: $\cos \frac{\alpha}{2} = -\sqrt{\frac{1+\cos}{2}}$ (因 $\frac{\alpha}{2}$ 在第二象限, 取负号)。(Half-angle formula: $\cos \frac{\alpha}{2} = -\sqrt{\frac{1+\cos}{2}}$ (take the negative sign because $\frac{\alpha}{2}$ is in the second quadrant).)
- 代入计算:(Substitute and calculate:)

$$\cos \frac{\alpha}{2} = -\sqrt{\frac{1+\left(-\frac{5}{13}\right)}{2}} = -\sqrt{\frac{\frac{8}{13}}{2}} = -\sqrt{\frac{4}{13}} = -\frac{2\sqrt{13}}{13}$$

- 故 C 正确。(Thus, C is correct.)

第 37 题

题目(Question): 设 $\cos \alpha = -\frac{1}{2}$, 且 $\alpha \in \left(\frac{\pi}{2}, \pi\right)$, 则 $\sin \frac{\alpha}{2} = ?$ (Let $\cos \alpha = -\frac{1}{2}$ and $\alpha \in \left(\frac{\pi}{2}, \pi\right)$. Then $\sin \frac{\alpha}{2} = ?$)

A $-\frac{\sqrt{3}}{2}$

B $\frac{\sqrt{3}}{2}$

C $-\frac{1}{2}$

D $\frac{1}{2}$

答案(Answer): B

解析(Explanation):

- 步骤 1: 确定 $\frac{\alpha}{2}$ 的象限 (Step 1: Determine the quadrant of $\frac{\alpha}{2}$):
 - 因 $\alpha \in \left(\frac{\pi}{2}, \pi\right)$, 两边除以 2 得 $\frac{\alpha}{2} \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ (第一象限), 第一象限中 $\sin \frac{\alpha}{2} > 0$ 。(Since $\alpha \in \left(\frac{\pi}{2}, \pi\right)$, divide both sides by 2 to get $\frac{\alpha}{2} \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ (first quadrant), where $\sin \frac{\alpha}{2} > 0$.)
- 步骤 2: 应用半角公式 (Step 2: Apply the half-angle formula):
 - 半角公式: $\sin \frac{\alpha}{2} = \sqrt{\frac{1-\cos \alpha}{2}}$ (因 $\frac{\alpha}{2}$ 在第一象限, 取正号)。(Half-angle formula: $\sin \frac{\alpha}{2} = \sqrt{\frac{1-\cos \alpha}{2}}$ (take the positive sign because $\frac{\alpha}{2}$ is in the first quadrant).)
 - 代入 $\cos \alpha = -\frac{1}{2}$ 计算:(Substitute $\cos \alpha = -\frac{1}{2}$ to calculate:)

$$\sin \frac{\alpha}{2} = \sqrt{\frac{1 - (-\frac{1}{2})}{2}} = \sqrt{\frac{\frac{3}{2}}{2}} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$

- 故 B 正确。(Thus, B is correct.)

第 38 题

题目(Question): 下列说法正确的是? (Which of the following statements is correct?)

A $\sin(\pi + \alpha) = \sin \alpha$

B $\cos(\pi + \alpha) = \cos \alpha$

C $\sin(-\alpha) = \sin \alpha$

D $\cos(-\alpha) = \cos \alpha$

答案(Answer): D

解析(Explanation):

- 三角函数诱导公式:(Trigonometric induction formulas:)
 - 选项 A: $\sin(\pi + \alpha) = -\sin \alpha$ (“奇变偶不变, 符号看象限”), A 错误; (Option A: $\sin(\pi + \alpha) = -\sin \alpha$ (“odd change even unchanged, sign depends on the quadrant”), A is incorrect;)
 - 选项 B: $\cos(\pi + \alpha) = -\cos \alpha$, B 错误; (Option B: $\cos(\pi + \alpha) = -\cos \alpha$, B is incorrect;)
 - 选项 C: $\sin(-\alpha) = -\sin \alpha$ (正弦函数是奇函数), C 错误; (Option C: $\sin(-\alpha) = -\sin \alpha$ (sine function is odd), C is incorrect;)
 - 选项 D: $\cos(-\alpha) = \cos \alpha$ (余弦函数是偶函数), D 正确。 (Option D: $\cos(-\alpha) = \cos \alpha$ (cosine function is even), D is correct.)

第 39 题

题目(Question): 关于正切函数(tangent function) $y = \tan x$, 下列说法正确的是? (About the tangent function $y = \tan x$, which of the following statements is correct?)



- A 值域(range)是 $[-1,1]$ (The range is $[-1,1]$)
- B 定义域(domain)是 $\mathbb{R}$ (The domain is $\mathbb{R}$)
- C 该函数是偶函数(The function is even)
- D 最小正周期(minimum positive period)是 π (The minimum positive period is π)

答案(Answer):D

解析(Explanation):

- 正切函数核心性质:(Core properties of the tangent function $y = \tan x$:)
 - 值域: $\mathbb{R}$ (全体实数), 而非 $[-1,1]$, A 错误; (Range: $\mathbb{R}$ (all real numbers), not $[-1,1]$, A is incorrect;)
 - 定义域: $x \neq \frac{\pi}{2} + k\pi$ ($k \in \mathbb{Z}$), 分母 $\cos x \neq 0$, 并非全体实数, B 错误; (Domain: $x \neq \frac{\pi}{2} + k\pi$ ($k \in \mathbb{Z}$), denominator $\cos x \neq 0$, not all real numbers, B is incorrect;)
 - 奇偶性: $\tan(-x) = -\tan x$, 是奇函数, 非偶函数, C 错误; (Parity: $\tan(-x) = -\tan x$, it is an odd function, not even, C is incorrect;)
 - 周期性:最小正周期为 π , D 正确。(Periodicity: The minimum positive period is π , D is correct.)

第 40 题

题目(Question): 数列 $-\frac{1}{5}, \frac{1}{7}, -\frac{1}{9}, \frac{1}{11}, \dots$ 的通项公式(general term formula)是? (What is the general term formula of the sequence $-\frac{1}{5}, \frac{1}{7}, -\frac{1}{9}, \frac{1}{11}, \dots$?)

- A $\frac{(-1)^{n-1}}{3n+2}$
- B $\frac{(-1)^n}{2n+3}$
- C $\frac{(-1)^{n-1}}{2n+3}$
- D $\frac{(-1)^n}{3n+2}$

答案(Answer):B

解析(Explanation):

- 分析数列规律:(Analyze the pattern of the sequence):
 - 符号规律:奇数项为负, 偶数项为正, 可用 $(-1)^n$ 表示($n = 1$ 时为 -1 , $n = 2$ 时为 1 , 符合符号规律); (Sign pattern: Odd terms are negative, even terms are positive, which can be expressed by $(-1)^n$ (when $n = 1$, it is -1 ; when $n = 2$, it is 1 , which matches the sign pattern);)
 - 分母规律:分母依次为 5, 7, 9, 11,..., 是公差为 2 的等差数列, 首项为 5。等差数列通项公式为 $a_n = a_1 + (n-1)d$, 代入得分母通项: $5 + (n-1) \times 2 = 2n + 3$ 。
(Denominator pattern: The denominators are 5, 7, 9, 11,... in sequence, which is an arithmetic sequence with common difference 2 and first term 5. The general term formula of the arithmetic sequence is $a_n = a_1 + (n-1)d$, substitute to get the denominator general term: $5 + (n-1) \times 2 = 2n + 3$.)
- 综上, 数列通项公式为 $a_n = \frac{(-1)^n}{2n+3}$, B 正确。(In summary, the general term formula of the sequence is $a_n = \frac{(-1)^n}{2n+3}$, so B is correct.)

第 41 题

题目(Question):若函数 $f(x) = 2 + \log_a(x-3)$ ($a > 0$ 且 $a \neq 1$)的图像过定点(fixed point)

P , 则 P 的坐标是? (If the graph of the function $f(x) = 2 + \log_a(x-3)$ ($a > 0$ and $a \neq 1$) passes through a fixed point P , what are the coordinates of P ?)

A (3, 1)

B (3, 0)

C (4, 0)



D (4, 2)

答案(Answer):D

解析(Explanation):

- 对数函数定点性质:对于 $\log_a N$ ($a > 0$ 且 $a \neq 1$), 无论 a 取何值, 当 $N = 1$ 时, $\log_a 1 = 0$ (与底数 a 无关). (Fixed point property of logarithmic functions: For $\log_a N$ ($a > 0$ and $a \neq 1$), no matter what value a takes, when $N = 1$, $\log_a 1 = 0$ (independent of the base a).)
- 令对数的真数 $x - 3 = 1$, 解得 $x = 4$ 。(Let the argument of the logarithm $x - 3 = 1$, solve to get $x = 4$.)
- 代入函数得 $f(4) = 2 + \log_a 1 = 2 + 0 = 2$, 故定点 P 的坐标为(4, 2), D 正确。
(Substitute into the function to get $f(4) = 2 + \log_a 1 = 2 + 0 = 2$, so the coordinates of the fixed point P are (4, 2), and D is correct.)

第 42 题

题目(Question): 抛物线 $y^2 = -x$ 的准线方程(directrix equation)是? (What is the directrix equation of the parabola $y^2 = -x$?)

A $x = \frac{1}{4}$

B $x = -\frac{1}{4}$

C $x = -\frac{1}{2}$

D $x = \frac{1}{2}$

答案(Answer):A

解析(Explanation):

- 抛物线标准形式及准线方程:对于抛物线 $y^2 = -2px$ ($p > 0$, 焦点在 x 轴负半轴), 准

线方程为 $x = \frac{p}{2}$ 。

(Standard form and directrix equation of the parabola: For the parabola $y^2 = -2px$

($p > 0$, focus on the negative semi-axis of the x-axis), the directrix equation is $x = \frac{p}{2}$.)

- 已知 $y^2 = -x$, 对比 $y^2 = -2px$ 得 $-2p = -1$, 即 $p = \frac{1}{2}$ 。

(Given $y^2 = -x$, compare with $y^2 = -2px$ to get $-2p = -1$, i.e., $p = \frac{1}{2}$.)

- 代入准线方程得 $x = \frac{p}{2} = \frac{1}{4}$, 故 A 正确。

(Substitute into the directrix equation to get $x = \frac{p}{2} = \frac{1}{4}$, so A is correct.)

第 43 题

题目(Question): 设椭圆(ellipse) $\frac{x^2}{4} + \frac{y^2}{2} = 1$ 的焦点为 F_1, F_2 , 点 P 在椭圆上, 则 $|PF_1| + |PF_2| = ?$

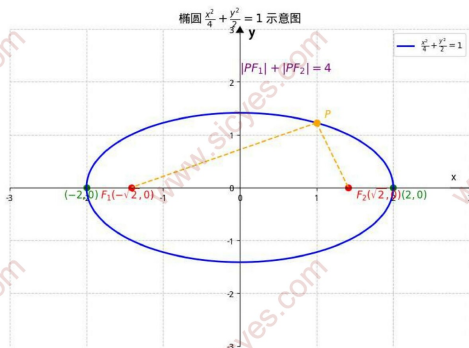
(Let F_1, F_2 be the foci of the ellipse $\frac{x^2}{4} + \frac{y^2}{2} = 1$, and point P be on the ellipse.

Then $|PF_1| + |PF_2| = ?$)

A 2 B 4 C 8 D 6

答案(Answer): B

解析(Explanation):





- 椭圆定义:平面内到两个焦点的距离之和为常数的点的轨迹,该常数为 $2a$ (a 为椭圆长半轴长)。(Definition of the ellipse: The locus of points in a plane where the sum of the distances to two foci is a constant, and this constant is $2a$ (a is the length of the semi-major axis of the ellipse).)
- 椭圆标准方程 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b > 0$), 其中 $a^2 = 4$, 故 $a = 2$ 。(Standard equation of the ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b > 0$), where $a^2 = 4$, so $a = 2$.)
- 由椭圆定义得 $|PF_1| + |PF_2| = 2a = 2 \times 2 = 4$, 故 B 正确。(From the ellipse definition, $|PF_1| + |PF_2| = 2a = 2 \times 2 = 4$, so B is correct.)

第 44 题

题目(Question):已知 $\overrightarrow{AB} = \vec{a} + 5\vec{b}$, $\overrightarrow{BC} = -2\vec{a} + 8\vec{b}$, $\overrightarrow{CD} = 3\vec{a} - 3\vec{b}$, 则下列说法正确的是? (Given $\overrightarrow{AB} = \vec{a} + 5\vec{b}$, $\overrightarrow{BC} = -2\vec{a} + 8\vec{b}$, $\overrightarrow{CD} = 3\vec{a} - 3\vec{b}$, which of the following statements is correct?)

- A A、C、D 三点共线(Points A, C, D are collinear)
- B A、B、D 三点共线(Points A, B, D are collinear)
- C B、C、D 三点共线(Points B, C, D are collinear)
- D A、B、C 三点共线(Points A, B, C are collinear)

答案(Answer):B

解析(Explanation):

- 三点共线的向量条件:若存在实数 λ ,使得两个有公共端点的向量共线(一个向量是另一个向量的实数倍),则三点共线。(Vector condition for three points to be collinear: If there exists a real number λ such that two vectors with a common endpoint are collinear (one vector is a real multiple of the other), then the three points are collinear.)

- 计算相关向量:(Calculate relevant vectors:)

○ 计算 $\overrightarrow{AD}$: $\overrightarrow{AD} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} = (\vec{a} + 5\vec{b}) + (-2\vec{a} + 8\vec{b}) + (3\vec{a} - 3\vec{b}) = 2\vec{a} + 10\vec{b} = 2(\vec{a} + 5\vec{b}) = 2\overrightarrow{AB}$ 。 (Calculate $\overrightarrow{AD}$: $\overrightarrow{AD} = \overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD} = (\vec{a} + 5\vec{b}) + (-2\vec{a} + 8\vec{b}) + (3\vec{a} - 3\vec{b}) = 2\vec{a} + 10\vec{b} = 2(\vec{a} + 5\vec{b}) = 2\overrightarrow{AB}$.)

- 分析共线性:(Analyze collinearity:)

○ $\overrightarrow{AD} = 2\overrightarrow{AB}$, 说明 $\overrightarrow{AD}$ 与 $\overrightarrow{AB}$ 共线, 且两向量有公共端点 A, 故 A、B、D 三点共线, B 正确。 ($\overrightarrow{AD} = 2\overrightarrow{AB}$ indicates that $\overrightarrow{AD}$ is collinear with $\overrightarrow{AB}$, and the two vectors have a common endpoint A, so points A, B, D are collinear, B is correct.)

第 45 题

题目(Question): 设直线 l 垂直于直线 $l_1: x + 2y + 4 = 0$, 且过直线 $l_2: x + y + 1 = 0$ 与 $l_3: 2x + y - 1 = 0$ 的交点(intersection point), 则直线 l 的方程是? (Suppose straight line l is perpendicular to line $l_1: x + 2y + 4 = 0$ and passes through the intersection point of lines $l_2: x + y + 1 = 0$ and $l_3: 2x + y - 1 = 0$. What is the equation of line l ?)

- A $2x - y - 7 = 0$
 B $2x + y - 5 = 0$
 C $x - 2y - 1 = 0$
 D $x + 2y - 3 = 0$ ($x + 2y - 3 = 0$)

答案(Answer):A

解析(Explanation):

- 步骤 1: 求 l_2 与 l_3 的交点(Step 1: Find the intersection of l_2 and l_3):
- 解方程组:(Solve the system of equations:)

$$\begin{cases} x + y = -1 \\ 2x + y = 1 \end{cases} \Rightarrow \begin{cases} x + y = -1 \\ x + y = 1 \end{cases}$$



- 两式相减得 $x = 2$, 代入 $x + y = -1$ 得 $y = -3$, 故交点为 $(2, -3)$ 。(Subtract the two equations to get $x = 2$, substitute into $x + y = -1$ to get $y = -3$, so the intersection point is $(2, -3)$.)
- 步骤 2:求直线 l 的斜率(Step 2: Find the slope of line l):
 - 直线 $l_1: x + 2y + 4 = 0$ 的斜率为 $k_1 = -\frac{1}{2}$ 。(The slope of line $l_1: x + 2y + 4 = 0$ is $k_1 = -\frac{1}{2}$.)
 - 因 $l \perp l_1$, 两直线垂直时斜率之积为 -1 , 故 l 的斜率 $k = 2$ ($-\frac{1}{2} \times 2 = -1$)。
(Since $l \perp l_1$, the product of slopes is -1 when two lines are perpendicular, so the slope of l is $k = 2$ ($-\frac{1}{2} \times 2 = -1$).)
- 步骤 3:求直线 l 的方程(Step 3: Find the equation of line l):
 - 用点斜式 $y - y_0 = k(x - x_0)$, 代入交点 $(2, -3)$ 和斜率 $k = 2$:(Use the point-slope form $y - y_0 = k(x - x_0)$, substitute the intersection point $(2, -3)$ and slope $k = 2$:)
 - $y - (-3) = 2(x - 2)$
 - 化简得:(Simplify to get:)
 - $y + 3 = 2x - 4 \Rightarrow 2x - y - 7 = 0$
- 故 A 正确。(Thus, A is correct.)

第 46 题

题目(Question):在复平面(complex plane)上, 复数 z 对应的点在直线 $x - y = 0$ 上, 若 z 是二次方程 $x^2 + mx + 4 = 0$ 的根(root), 则 $m = ?$ (On the complex plane, the point corresponding to the complex number z is on the line $x - y = 0$. If z is a root of the quadratic equation $x^2 + mx + 4 = 0$, then $m = ?$)

A $\sqrt{2}$ 或 $2\sqrt{2}$

B $2\sqrt{2}$ 或 $-2\sqrt{2}$

C $-\sqrt{2}$ 或 $-2\sqrt{2}$

D $\sqrt{2}$ 或 $-\sqrt{2}$

答案(Answer):B

解析(Explanation):

- 步骤 1: 设复数 z 的形式 (Step 1: Set the form of complex number z):
 - 复平面上直线 $x - y = 0$ 即 $x = y$, 故设 $z = a + ai$ ($a \in \mathbb{R}$, 实部=虚部= a)。(The line $x - y = 0$ on the complex plane is $x = y$, so let $z = a + ai$ ($a \in \mathbb{R}$, real part = imaginary part = a .)
- 步骤 2: 代入二次方程求解 (Step 2: Substitute into the quadratic equation to solve):
 - 因 z 是方程 $x^2 + mx + 4 = 0$ 的根, 代入 $z = a + ai$: (Since z is a root of the equation $x^2 + mx + 4 = 0$, substitute $z = a + ai$:)
 $(a + ai)^2 + m(a + ai) + 4 = 0$
 - 展开并整理: (Expand and simplify):
 $a^2 + 2a^2i - a^2 + ma + mai + 4 = 0$
 $(ma + 4) + (2a^2 + ma)i = 0$
 - 复数为 0 的充要条件: 实部和虚部分别为 0, 故: (Necessary and sufficient condition for a complex number to be 0: The real part and imaginary part are 0 respectively, so:)

$$\begin{cases} ma + 4 = 0 & (1) \\ 2a^2 + ma = 0 & (2) \end{cases}$$

- 步骤 3: 解方程组 (Step 3: Solve the system of equations):
 - 由(1)得 $ma = -4$, 代入(2): (From (1), $ma = -4$, substitute into (2):)
 $2a^2 - 4 = 0 \Rightarrow a^2 = 2 \Rightarrow a = \pm\sqrt{2}$
 - 代入 $ma = -4$: 当 $a = \sqrt{2}$ 时, $m = -\frac{4}{\sqrt{2}} = -2\sqrt{2}$; 当 $a = -\sqrt{2}$ 时, $m = \frac{4}{\sqrt{2}} = 2\sqrt{2}$ 。(Substitute into $ma = -4$: When $a = \sqrt{2}$, $m = -\frac{4}{\sqrt{2}} = -2\sqrt{2}$; when $a = -\sqrt{2}$, $m = \frac{4}{\sqrt{2}} = 2\sqrt{2}$.)



- 故 $m = 2\sqrt{2}$ 或 $-2\sqrt{2}$, B 正确。(Thus, $m = 2\sqrt{2}$ or $-2\sqrt{2}$, so B is correct.)

第 47 题

题目(Question): 在数列 $\{a_n\}$ 中, $a_1 = 1$, 且 $\frac{1}{a_n} + \frac{2}{a_{n+1}} = 0$ ($n = 1, 2, \dots$)。数列 $\{b_n\}$ 满足 $\{b_n\} = |a_n|$ ($n = 1, 2, \dots$) , 则 b_n 的前 n 项和(sum of the first n terms) $S_n = ?$ (In the sequence $\{a_n\}$, $a_1 = 1$ and $\frac{1}{a_n} + \frac{2}{a_{n+1}} = 0$ ($n = 1, 2, \dots$). The sequence $\{b_n\}$ satisfies $b_n = |a_n|$ ($n = 1, 2, \dots$). Then the sum of the first n terms of b_n is $S_n = ?$)

A $\frac{1+2^n}{3}$

B $\frac{|1-(-2)^n|}{3}$

C $2^n - 1$

D $(-2)^n - 1$

答案(Answer): C

解析(Explanation):

- 步骤 1: 求数列 $\{a_n\}$ 的通项公式(Step 1: Find the general term formula of a_n):
 - 由 $\frac{1}{a_n} + \frac{2}{a_{n+1}} = 0$, 移项得 $\frac{2}{a_{n+1}} = -\frac{1}{a_n}$, 即 $\frac{a_{n+1}}{a_n} = -2$ 。(From $\frac{1}{a_n} + \frac{2}{a_{n+1}} = 0$, rearrange to get $\frac{2}{a_{n+1}} = -\frac{1}{a_n}$, i.e., $\frac{a_{n+1}}{a_n} = -2$.)
 - 因此, $\{a_n\}$ 是首项 $a_1 = 1$ 、公比 $q = -2$ 的等比数列。(Thus, $\{a_n\}$ is a geometric sequence with first term $a_1 = 1$ and common ratio $q = -2$.)
 - 等比数列通项公式: $a_n = a_1 q^{n-1} = (-2)^{n-1}$ 。(General term formula of geometric sequence: $a_n = a_1 q^{n-1} = (-2)^{n-1}$.)
- 步骤 2: 求数列 $\{b_n\}$ 的前 n 项和(Step 2: Find the sum of the first n terms of $\{b_n\}$):
 - 由 $b_n = |a_n|$, 得 $b_n = |(-2)^{n-1}| = 2^{n-1}$ (绝对值消去负号, 数列变为正项等比数列)。(From $b_n = |a_n|$, we get $b_n = |(-2)^{n-1}| = 2^{n-1}$ (the absolute value eliminates the

negative sign, and the sequence becomes a positive-term geometric sequence.)

- o $\{b_n\}$ 的首项 $b_1 = 1$ 、公比 $q' = 2$, 等比数列前 n 项和公式为 $S_n = \frac{b_1(1-q'^n)}{1-q'}$ 。

($\{b_n\}$ has first term $b_1 = 1$ and common ratio $q' = 2$. The formula for the sum of the first n terms of a geometric sequence is $S_n = \frac{b_1(1-q'^n)}{1-q'}$.)

- o 代入计算: $S_n = \frac{1 \times (1-2^n)}{1-2} = 2^n - 1$ 。(Substitute to calculate: $S_n = \frac{1 \times (1-2^n)}{1-2} = 2^n - 1$.)

- 故 C 正确。(Thus, C is correct.)

第 48 题

题目(Question):为了解某地区高中生和大学生对跑步 APP 的使用偏好, 随机选取 200 名高中生和 80 名大学生, 记录其喜欢的 APP 如下表。假设两类学生的偏好相互独立(independent), 从每组中随机选取 1 人, 两人喜欢的 APP 不同的概率(probability)是? (To understand the preference for running apps among high school and college students in a region, 200 high school students and 80 college students were randomly selected, and their favorite apps are recorded in the table below. Assuming the preferences of the two groups are independent, if one person is randomly selected from each group, what is the probability that their favorite apps are different?)

	App A	App B	App C	App D
高中生(High school students)	80	40	60	20
大学生(College students)	30	20	20	10

A $\frac{23}{80}$

B $\frac{57}{80}$

C $\frac{9}{20}$

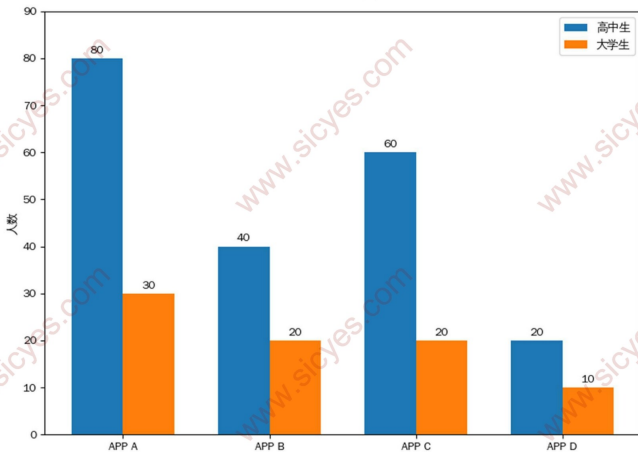
D $\frac{11}{20} \left(\frac{11}{20} \right)$



答案(Answer):B

解析(Explanation):

不同APP的学生使用人数对比



解题思路:利用“对立事件概率”计算, 即“APP不同的概率 = 1 - APP相同的概率”。(Problem-

solving idea: Calculate using the “probability of complementary events”: Probability of different

apps = 1 - Probability of the same app.)

- 步骤 1:计算每组学生选择各 APP 的概率(Step 1: Calculate the probability of each group choosing each app):

○ 高中生选择各 APP 的概率: App A = $\frac{80}{200} = \frac{2}{5}$, App B = $\frac{40}{200} = \frac{1}{5}$, App C = $\frac{60}{200} = \frac{3}{10}$,

App D = $\frac{20}{200} = \frac{1}{10}$ 。(Probability of high school students choosing each app: App A =

$\frac{80}{200} = \frac{2}{5}$, App B = $\frac{40}{200} = \frac{1}{5}$, App C = $\frac{60}{200} = \frac{3}{10}$, App D = $\frac{20}{200} = \frac{1}{10}$.)

- 大学生选择各 APP 的概率: App A = $\frac{30}{80} = \frac{3}{8}$, App B = $\frac{20}{80} = \frac{1}{4}$, App C = $\frac{20}{80} = \frac{1}{4}$, App

D = $\frac{10}{80} = \frac{1}{8}$ 。(Probability of college students choosing each app: App A = $\frac{30}{80} = \frac{3}{8}$,

App B = $\frac{20}{80} = \frac{1}{4}$, App C = $\frac{20}{80} = \frac{1}{4}$, App D = $\frac{10}{80} = \frac{1}{8}$.)

- 步骤 2: 计算“APP 相同”的概率(Step 2: Calculate the probability of “the same app”):

- 相同 APP 的概率 = (同选 A)+(同选 B)+(同选 C)+(同选 D)。(Probability of the same app = (both choose A) + (both choose B) + (both choose C) + (both choose D).)

- 代入计算:(Substitute to calculate:)

$$\left(\frac{2}{5} \times \frac{3}{8}\right) + \left(\frac{1}{5} \times \frac{1}{4}\right) + \left(\frac{3}{10} \times \frac{1}{4}\right) + \left(\frac{1}{10} \times \frac{1}{8}\right) = \frac{6}{40} + \frac{2}{40} + \frac{3}{40} + \frac{0.5}{40} = \frac{23}{80}$$

- 步骤 3: 计算“APP 不同”的概率(Step 3: Calculate the probability of “different apps”):

- 不同 APP 的概率 = $1 - \frac{23}{80} = \frac{57}{80}$ 。(Probability of different apps = $1 - \frac{23}{80} = \frac{57}{80}$.)

- 故 B 正确。(Thus, B is correct.)